BandLink firmware architecture

Layered view of the STM32U5G9 firmware, from the app screens down to the silicon. FreeRTOS with preemptive priorities: interrupts wake the owning task, tasks pass data through queues, and tickless idle drops into Stop 2 whenever every task is blocked. Each layer only calls the layer below it.

S1 Stretch 1 S2 Stretch 2 Untagged = MVP

Application

What the wearer sees and does

- **Finder screen** — Arrow + distance to the selected friend; switches to UWB distance when in range
- **Group screen** — Paired friends, last-seen age, battery, signal
- **Pairing flow** — Crown long-press → tap → confirm → key saved
- **Watch face + settings** — Time from GPS, brightness, mode, unpair
- **Mode controller** — Standby / Typical / Finding: sets TX interval, screen, IMU rate, UWB on/off

UI framework

LVGL 9 (MIT, C) · used only by the Application layer

- **Widgets + screens** — `lv_arc` ring, `lv_label` distance, rotated `lv_image` arrow, `lv_list` group
- **Styles + theme** — Dark theme for AMOLED power, round 466×466 layout, fonts in flash
- **Input devices** S1 — PEC06 → `LV_INDEV_TYPE_ENCODER` + focus group; touch → pointer indev
- **Timers + animation** — `lv_timer_handler()` in the ui task; `lv_tick_inc()` from the FreeRTOS tick hook
- **Rendering** — Two partial buffers (466×46 RGB565 ≈ 43 KB each) in SRAM; NeoChrom (NemaGFX) draw unit S1, DMA2D / software fallback
- **Display port** — `flush_cb` starts OCTOSPI DMA on TE; DMA-complete ISR calls `lv_display_flush_ready()`; round dirty areas to even x/y (CO5300 alignment, confirm)

Services

Hardware-independent logic

- **Link layer** — Packet codec, GPS-timed TDMA slots, dedupe, relay queue S2
- **Crypto** — AES-128-CCM encrypt + MIC, nonce = ID ‖ seq; uses HW AES
- **Navigation** — Peer table, haversine distance, initial bearing, heading from quaternion
- **Pairing service** — Group key + device ID exchange over NFC, writes key store
- **Ranging service** S2 — DS-TWR at 10 Hz when a peer is within ~30 m
- **Battery service** — SOC, charge state, low-battery mode
- **Haptics service** — Named patterns: paired, friend found, low battery

RTOS

FreeRTOS kernel and power

- **FreeRTOS kernel** — 7 tasks, preemptive priorities; X-CUBE-FREERTOS via CubeMX
- **Task notifications + queues** — ISRs wake tasks with `vTaskNotifyGiveFromISR()`; services pass data through queues
- **Mutexes + software timers** — Bus mutexes (priority inheritance); timers for TX slots, LVGL tick, polling
- **Tickless idle** — Tick from LPTIM1; all tasks blocked → Stop 2; HAL timebase on TIM6
- **Power manager** — Load interlocks (no NFC field during LoRa TX / UWB), rail and peripheral gating
- **Persistent store** — Wear-levelled flash: keys, IMU calibration, settings

Device drivers

One per chip

- **SX1262** — Semtech sx126x driver + BUSY wait, DIO1 IRQ
- **MAX-M10S** — UBX parser + config (rate, constellations, power-save); PPS capture
- **PN7150** — NCI host stack (NXP MCU examples), P2P / card emulation
- **BNO085** — CEVA sh2 library over SHTP; rotation vector, motion reports
- **CO5300 AMOLED** S1 — QSPI command set, TE-synced flush, brightness
- **DW3000** S2 — Qorvo DW3000 API, deep-sleep / wake
- **DRV2605L · MAX17048 · BQ25180** S1 — Small register drivers on I2C2
- **PEC06 crown** S1 — Timer encoder count + debounced push

HAL / BSP

STM32CubeU5

- **Buses** — SPI1–3, I2C1–2, USART1–2, OCTOSPI1
- **DMA** — GPDMA for SPI, QSPI frame flush, UART RX
- **Timers + EXTI** — TIM2 PPS capture, TIM3 encoder, LPTIM tick, 9 EXTI lines
- **Security blocks** — AES, RNG, PKA, hash
- **Graphics** S1 — NeoChrom GPU2D, DMA2D
- **System** — Clocks, PWR/SMPS, RTC, flash, IWDG

Hardware

Per bracelet

- **STM32U5G9** — Cortex-M33, ~3 MB SRAM
- **Radios** — SX1262 · PN7150 · DWM3000 S2
- **Sensors** — MAX-M10S GNSS · BNO085 IMU
- **Interface** — AMOLED · PEC06 · DRV2605L + LRA S1
- **Power** — BQ25180 · TPS63802 · MAX17048 · 500 mAh LiPo

CROSS-CUTTING

- **Security** — Group key never leaves the device after pairing; RDP level 1 on release builds; RNG-seeded nonces

Power policy

Every service declares its wake needs; power manager picks the deepest allowed stop mode

Logging + console

Ring-buffer log, UART console on the pogo dock, SEGGER RTT over SWD in dev

Firmware update

ST system bootloader over USART1 on the dock; BNO085 updated through BOOTN

Testing

Services build on the host (no HAL) for unit tests: codec, crypto, nav math

Rule: Application and Services never include a HAL header. Hardware access goes through a driver interface, so the same service code runs on the Nucleo, the custom PCB, and a PC test build.

RTOS tasks

Higher-priority tasks preempt lower ones, so a long UI redraw never delays radio work. Each task blocks on a notification or queue and costs no CPU while waiting.

TASK	PRIORITY	WAKES ON	OWNS	STACK (START)
`radio`	6 (highest)	DIO1 IRQ, TDMA slot timer	SX1262 driver, link layer, crypto, SPI1	2 KB
`ranging` S2	5	UWB IRQ, 100 ms timer in Finding mode	DW3000 driver, ranging service, SPI2	2 KB
`position`	4	USART2 DMA idle-line, PPS capture	UBX parser, navigation, peer table	2 KB
`sensor`	3	BNO085 H_INTN	sh2 library, heading, wrist-raise, SPI3	2 KB
`pairing`	3	Crown long-press, PN7150 IRQ	NCI host, pairing service, I2C1	3 KB
`ui`	2	LVGL timer, display TE, queue from services	LVGL (only task allowed to call it), display driver, haptics	6 KB
`system`	1 (lowest)	1 s timer, ALRT, charger INT	Battery service, power manager, watchdog kick, log flush	1.5 KB

RTOS rules

- ISRs do the minimum: timestamp, start DMA, notify the task. Only `...FromISR()` APIs inside interrupts.
- One task owns each bus; shared I2C2 goes through a mutex with priority inheritance.
- Only the ui task calls LVGL. Other tasks send it display updates through a queue.
- Timing-critical radio actions stay in hardware: the slot-timer ISR writes SetTx, and UWB replies use the DW3000's delayed-TX.
- Debug builds keep stack-overflow checking, `configASSERT`, and stack high-water checks on.

One packet's path: a friend's position reaches the screen

1	**SX1262**	Packet ends; DIO1 fires the EXTI interrupt.
2	**ISR → radio task**	ISR calls `vTaskNotifyGiveFromISR()`; the core wakes from Stop 2 and the radio task, the highest priority, runs and reads the FIFO over SPI1.
3	**link + crypto**	Checks seq against the dedupe table, verifies the AES-CCM MIC with the group key, decrypts. Bad MIC → dropped.
4	**link (S2)**	If relay is on and the hop count allows, queues a re-broadcast for the next free slot.
5	**navigation**	Updates the peer table; computes distance and bearing from this bracelet's last GPS fix.
6	**sensor**	Latest BNO085 heading gives the arrow angle: bearing − heading.
7	**ui**	Navigation posts the update to the ui task's queue; the ui task invalidates the arrow and distance widgets, and LVGL redraws only those areas on the next TE pulse.
8	**haptics**	If this friend just came back into range, plays the "found" pattern.